



Solar PV Potential Analysis for CLV

JACK WILLIAMS

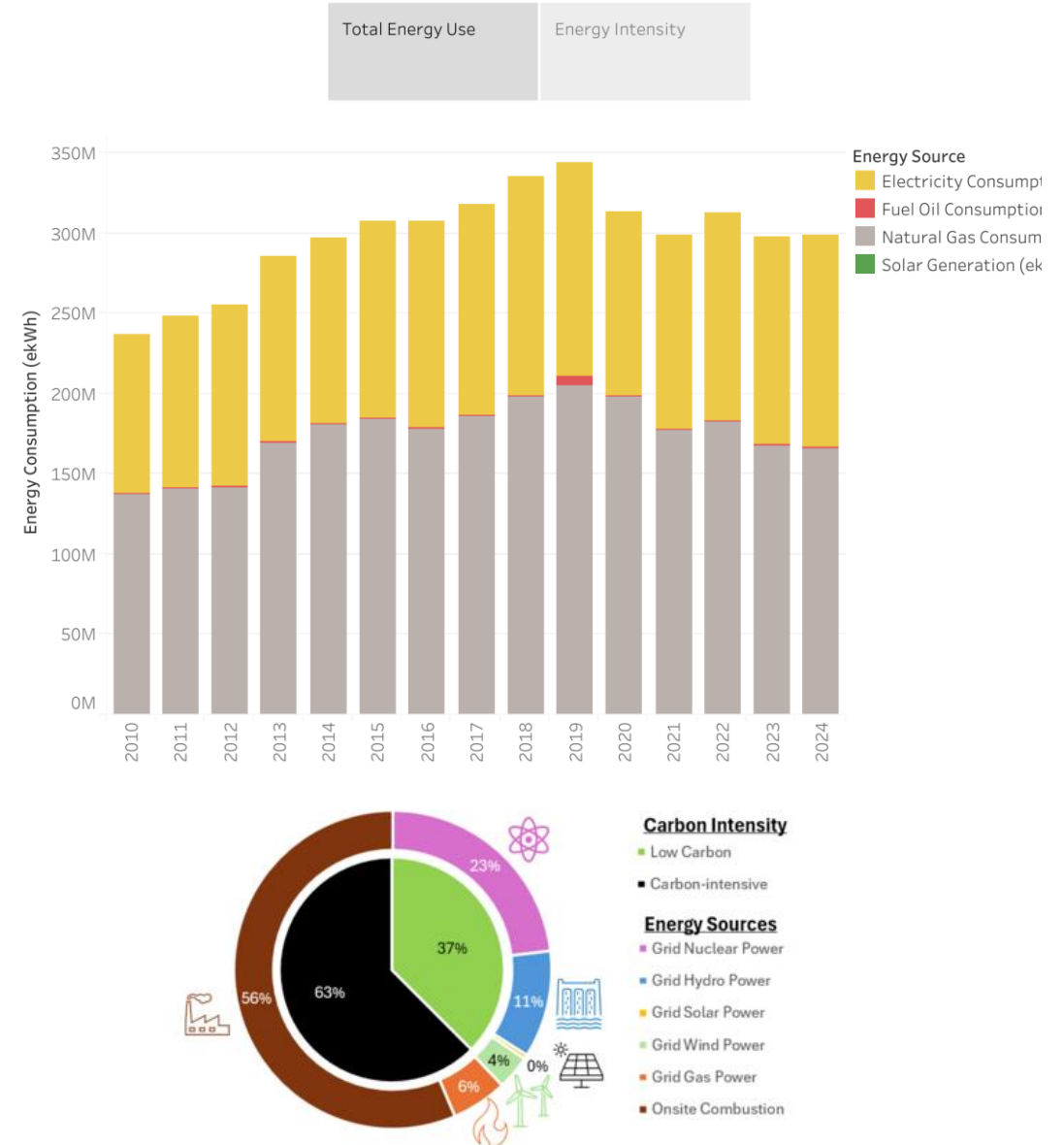
KAILIN ZHANG

WILLIAM ZHANG

Overview of the Energy Consumption Problem

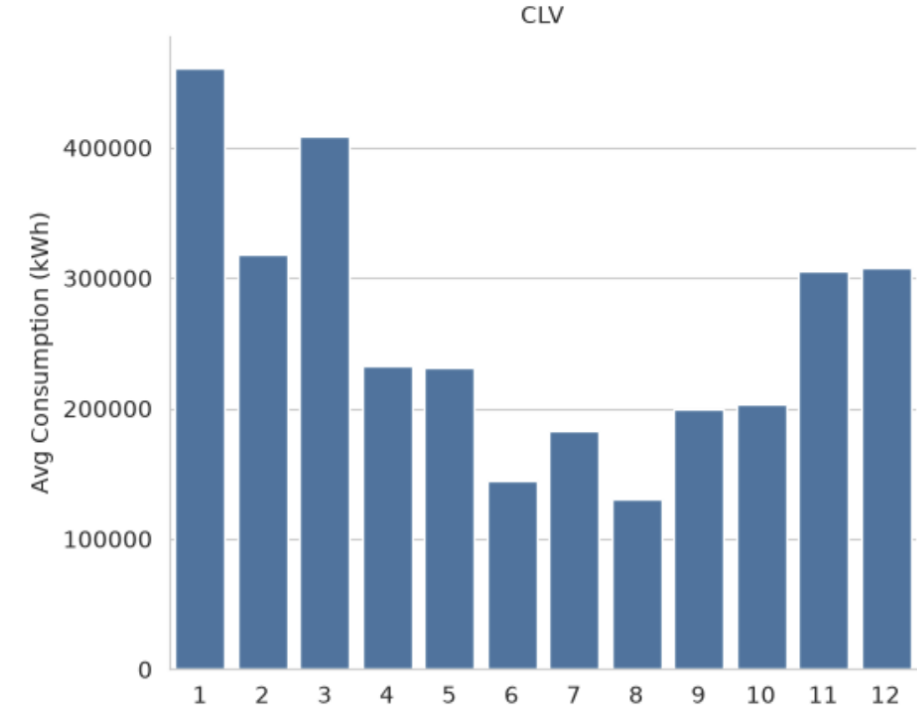
- Waterloo has a high climate variability
- 63% of UW energy use is from non-renewable sources (2015)
- Electricity consumption is a significant portion of the campus energy use

Campus Energy Use



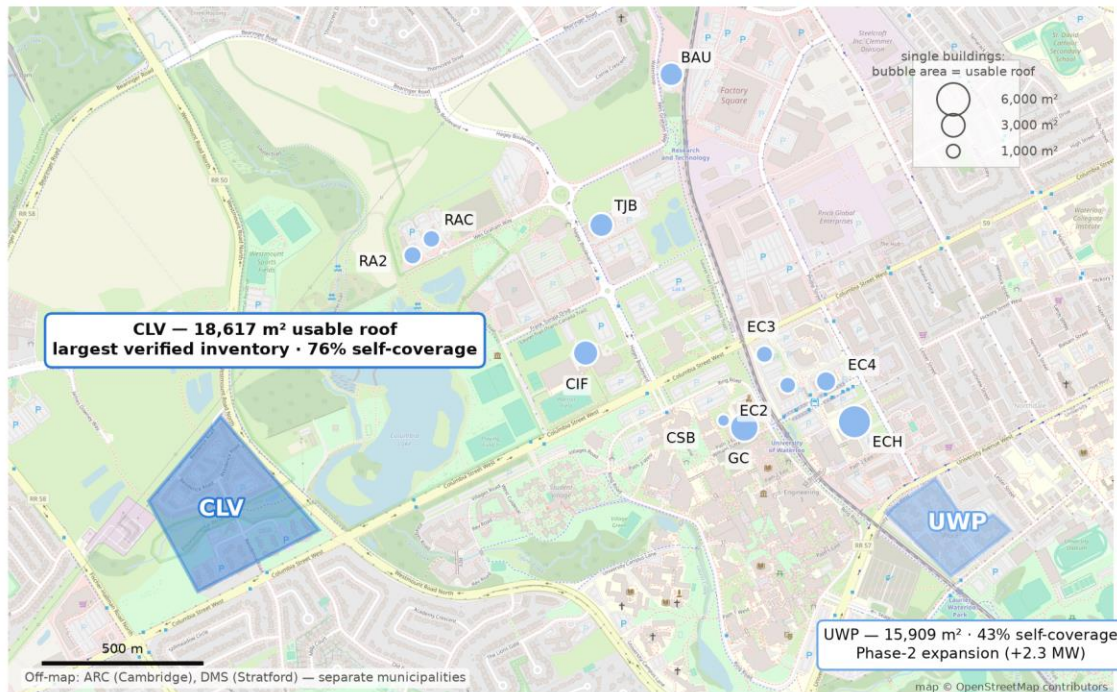
The Renewable Opportunity for CLV

- Relatively large area of campus
- Moderately high monthly energy consumption
- All units are only 2 storeys
- Close proximity to weather station
- Mostly consists of grad students
- 76% self coverage (total electricity consumption that could be generated by solar panels)
- Individual units are connected in groups of 6 and housing blocks are closely spaced

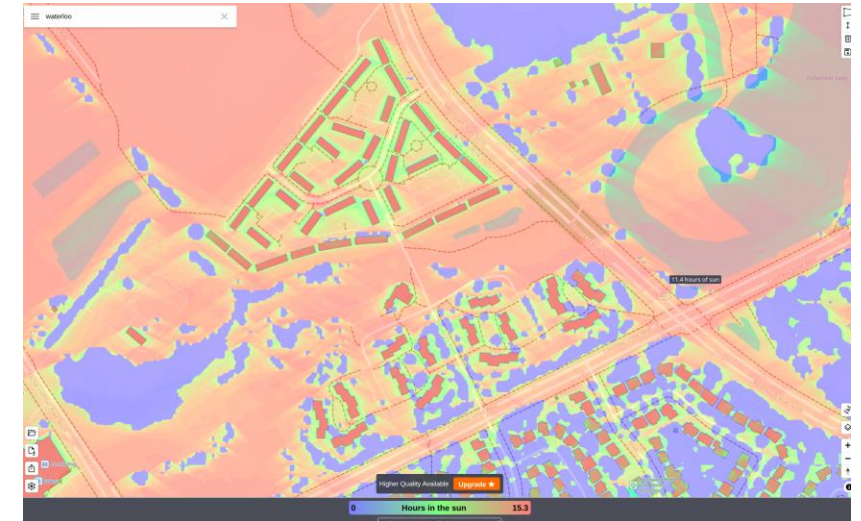


Shade Map and Site Suitability Mapping

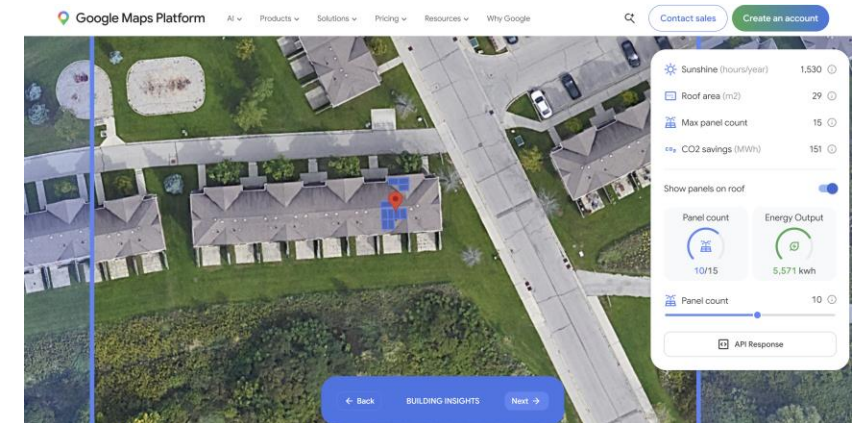
Every building in the challenge dataset, measured — Google Solar API over verified boundaries



Total area from QGIS: 19,883 m²

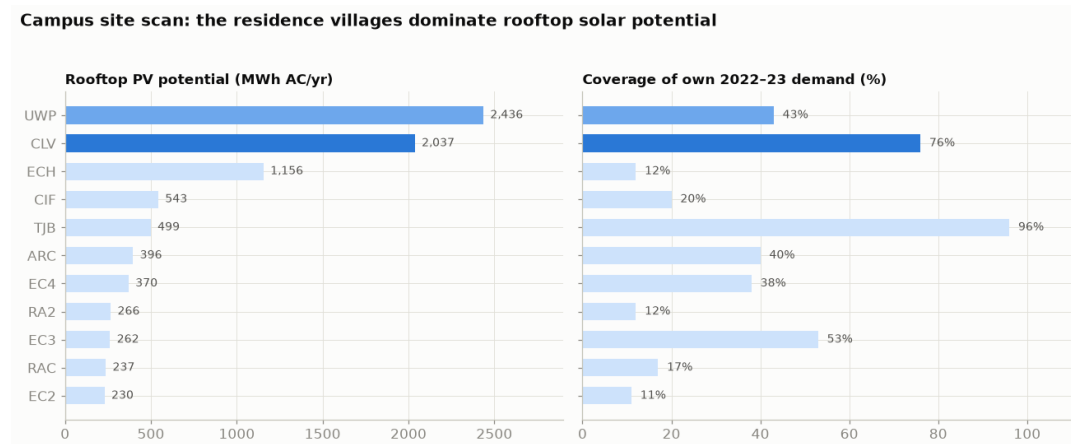
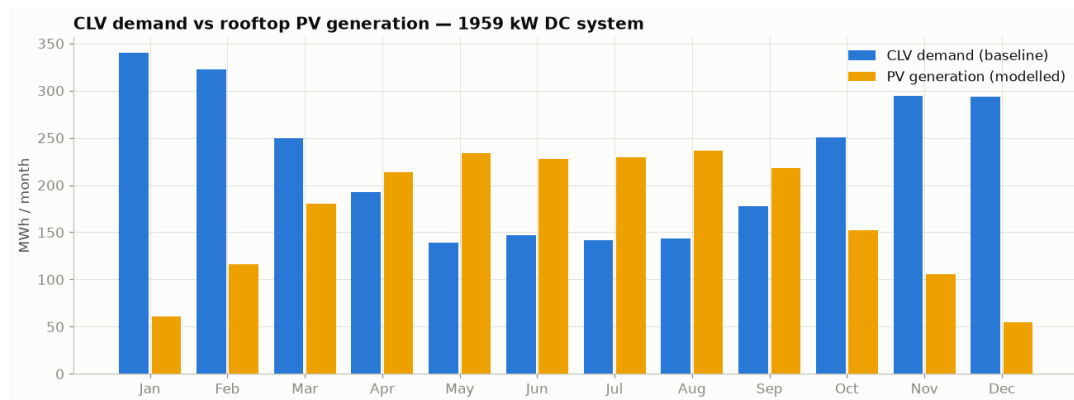


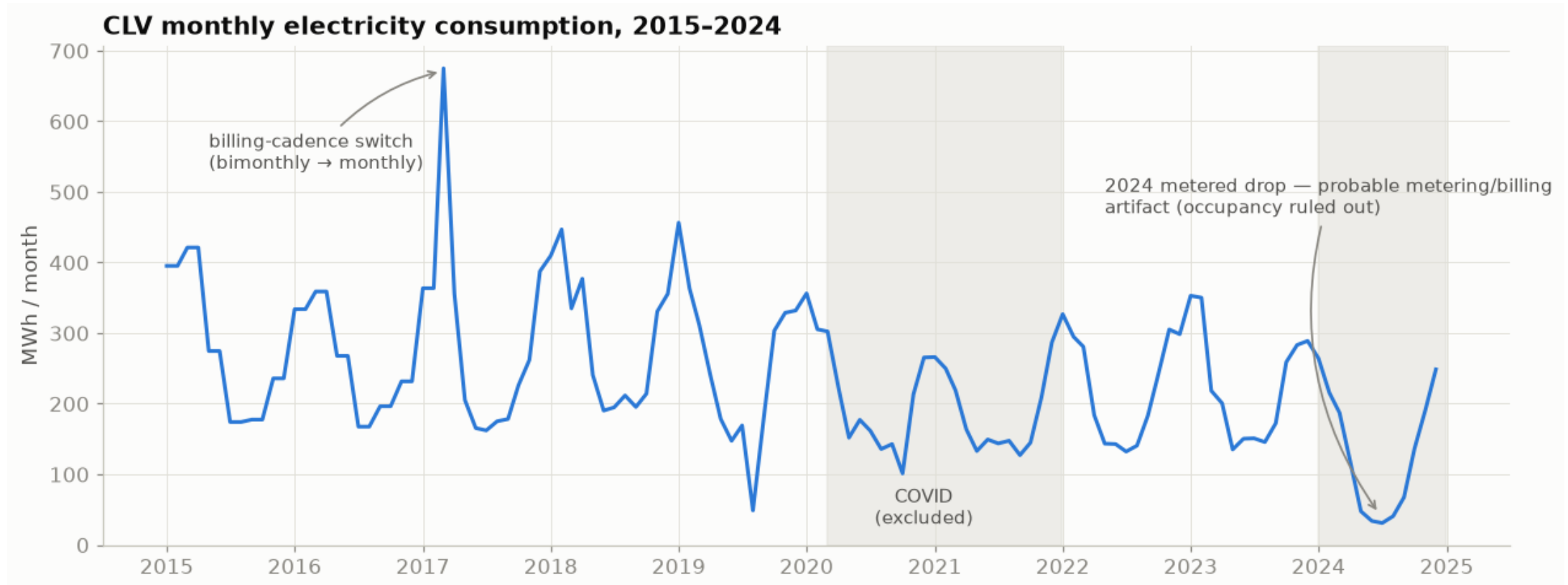
Summer Solstice: 14 hours
Winter Solstice: 8 hours



CLV Energy Consumption

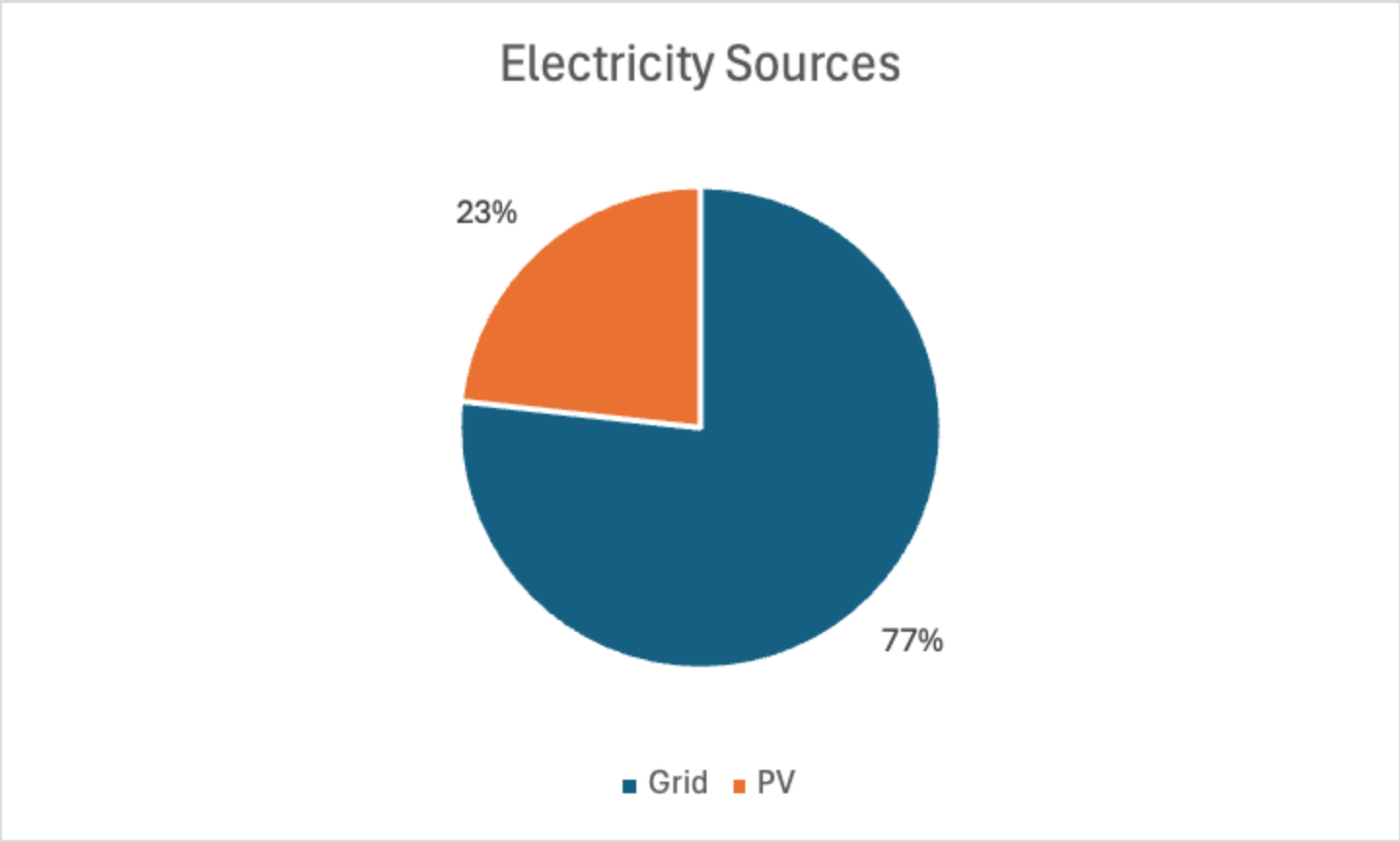
18 panels per dwelling unit
4,897 panels across 269 units





CLV Energy Consumption

Grey bands = excluded from baseline (COVID: real demand, abnormal period · 2024: metering artifact, occupancy verified). Baseline 2022–23: 2.69 GWh/yr.





Impact

2030

23% Less
Grid Dependence

2040

Cash Flow
Positive

2050

1M kWh
\$2M Saved

Total GHG reduction: 350 tCO₂e

Economic Snapshot

Initial Cost \$600,000
Annual Energy Savings \$60,000 - \$90,000
Simple Payback Period 10 Yrs

Payback Cash Flow (Payback Period = 10.5 years)

